Supplementary Material

Article OmicsXplore: A Fully Cloud-Based Platform for Automated, End-to-End Bioinformatics Analysis Applied to RNA Sequencing of Idiopathic Pulmonary Fibrosis Lung Tissue

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Contents

S1 Module inventory, versions and licenses (Table S1)

S2 Platform implementation notes

S3 Case-study data, settings and results (Tables S2 to S9, Figs. S1 to S5)

S4 Independent reproduction of the differential expression result

S5 Pseudocode of the pipeline driver (Algorithms 1 and 2)

S6 Supplementary Data S1 and S2

References

S1 Module inventory, versions and licenses

Table S1 lists every third-party engine and external resource that the RNA-seq workflow calls, with the version used in the case study, the license under which it is distributed and the publication to cite. Each engine runs unmodified, under its original license, on the platform's cloud infrastructure, and users receive the outputs of their own jobs, not the software itself. Licenses were taken from the repository or package record of each project. Web services are called under the terms of their providers. The gene set library used for enrichment is derived from the Gene Ontology, whose data are distributed under the Creative Commons Attribution 4.0 license [1].

Table S1. Third-party engines and resources called by the OmicsXplore RNA-seq workflow, with the versions used in the case study, their licenses and the references to cite. Module numbers follow Table II of the main article. n.r., version not reported.

Module	Engine or resource	Version	License or terms	Reference
1 Upload	SRA Toolkit, fasterq-dump	3.0.3	Public domain (United States Government work)	[2]
2 and 4, Pre-trim QC and Post-trim QC	FastQC	0.12.1	GPL-3.0	[3]
2 and 4	MultiQC	1.35	GPL-3.0	[4]
3 Trimming	Trimmomatic, with the TruSeq2 paired-end adapter file	0.39	GPL-3.0. Illumina adapter sequences remain the property of Illumina and are distributed with Trimmomatic by permission	[5]
5 Alignment	STAR	2.7.11b	MIT	[6]
5	UCSC hg38 genome (hg38.fa) and knownGene annotation (hg38.knownGene.gtf)	hg38	Free to use under the UCSC data terms	[7]
6 Pre-quantification	SAMtools	1.22.1	MIT/Expat	[8], [9]
7 Quantification	HTSeq-count	2.0.5	GPL-3.0	[10], [11]
7	featureCounts (Subread), offered as an alternative and not used in the case study	2.0.6	GPL-3.0	[12]
8 Annotation	g:Profiler g:Convert	Web service	Terms of the provider	[13]
9 DEG analysis	PyDESeq2	0.5.4	MIT	[14], [15]
9	MyGene.info	Web service	Terms of the provider	[16]
9	GSEApy	1.3.1	BSD-3-Clause	[17]
9	Enrichr, with the GO_Biological_Process_2023 library	Web service, 2023 library	Terms of the provider. Gene Ontology data CC BY 4.0	[1], [18]–[21]
9	scikit-learn, SciPy, pandas and seaborn	n.r.	BSD-3-Clause	[22]–[25]
9	matplotlib	n.r.	Matplotlib license, based on the PSF license	[26]
10 Gene Splicing	splicekit	0.8.1	GPL-3.0	[27]
10	edgeR	n.r.	GPL (2 or later)	[28], [29]
10 and 11	STAR, re-run with module-specific settings	2.7.11b	MIT	[6]
11 Gene fusion Detection	Arriba, with its blacklist, known fusion and protein domain files	2.5.0	MIT. The draw_fusions.R plotting script is GPL-3.0	[30]

S2 Platform implementation notes

S2.1 Architecture

OmicsXplore has a client and server design that follows SilicoXplore [31]. The backend is written in Python 3 with the Flask framework and exposes a REST interface for job submission, status queries, result retrieval and file management. Each analysis module is registered as a backend service that declares its input schema, validation rules, default parameters and the engine it wraps, and the browser form of the module is generated from this declaration. The frontend is a single-page application written in React with TypeScript. Inputs are checked against the declaration before a job is queued. A malformed accession list, a missing file or a metadata table whose sample

names do not match the count matrix is therefore reported at submission and costs no compute time.

S2.2 Job execution

Jobs are queued and dispatched by a scheduler developed in house. The scheduler records the state of every stage of every job in a database and starts a stage when processor cores become free. Each process receives a bounded thread pool for numerical libraries such as BLAS, OpenMP and joblib, so concurrent jobs do not oversubscribe a server. Every stage reads its input from, and writes its output to, one directory identified by the job identifier. A partial run can therefore resume from the last completed stage. A stage started from the interface and the same stage run by the pipeline driver call the same service function with the same arguments (Section S5). Third-party engines run in versioned software environments on the infrastructure of a commercial cloud provider, and the engine versions recorded with a job are those of the environment in which it ran.

S2.3 Storage and access

Uploaded reads, downloaded runs, intermediate files, logs and results are kept in the project workspace in server-side storage. Users can open only the projects created under their own login. Files remain available for download from the result pages, and the output of one module can be selected as the input of the next without being downloaded.

S2.4 The job record

Every job keeps a record with the job identifier, the input datasets, the complete parameter set as submitted, the engine versions, the start time, the status and run time of each stage, the logs and the output files. The pipeline driver also writes a run report in JSON format that lists each stage with its outcome and a short summary (Algorithm 1). The summary gives values such as the number of samples processed, the alignment rate or the number of significant features. The parameter set appears on the result page exactly as it was submitted. The settings reported in Section III of the main article and in Table S4 were read from these records.

S2.5 Result presentation

Each module returns summary tables and plots on its result page, together with links to the raw outputs. The quality control modules return an aggregated MultiQC report. The DEG analysis module returns the full and filtered result tables, separate lists of features with positive and negative fold changes, the PCA, volcano, MA and clustered heatmap plots, and the enrichment tables and plots. The Gene Splicing module returns the splicekit HTML report with its tables. The Gene fusion Detection module returns the Arriba fusion table, the discarded candidates and one PDF of visualizations per sample.

S3 Case-study data, settings and results

S3.1 Samples

Table S2 lists the 39 runs of GSE92592 analyzed in this study [32], [33]. All libraries were prepared from lung tissue without strand selection and sequenced on an Illumina HiSeq 2000 as 2 × 101 bp read pairs. Group labels were taken from the GEO sample records and entered into the DEG analysis module as the condition column of the metadata table.

Table S2. Samples of GSE92592 analyzed in the case study. Read pairs are the FastQC sequence counts of the forward files, and bases are taken from the SRA run table.

Run	GEO sample	BioSample	SRA experiment	Group	Read pairs (million)	Bases (billion)
SRR5120902	GSM2433098	SAMN06165919	SRX2436128	IPF	60.16	12.15
SRR5120903	GSM2433099	SAMN06165918	SRX2436129	IPF	53.19	10.74
SRR5120904	GSM2433100	SAMN06165917	SRX2436130	IPF	33.10	6.69
SRR5120905	GSM2433101	SAMN06165916	SRX2436131	IPF	49.80	10.06
SRR5120906	GSM2433102	SAMN06165915	SRX2436132	IPF	60.36	12.19
SRR5120907	GSM2433103	SAMN06165914	SRX2436133	IPF	34.69	7.01
SRR5120908	GSM2433104	SAMN06165913	SRX2436134	IPF	42.94	8.67
SRR5120909	GSM2433105	SAMN06165912	SRX2436135	IPF	45.67	9.23
SRR5120910	GSM2433106	SAMN06165911	SRX2436136	IPF	54.46	11.00
SRR5120911	GSM2433107	SAMN06165910	SRX2436137	IPF	57.04	11.52
SRR5120912	GSM2433108	SAMN06165909	SRX2436138	IPF	48.15	9.73
SRR5120913	GSM2433109	SAMN06165908	SRX2436139	IPF	39.52	7.98
SRR5120914	GSM2433110	SAMN06165907	SRX2436140	IPF	43.91	8.87
SRR5120915	GSM2433111	SAMN06165906	SRX2436141	IPF	42.69	8.62
SRR5120916	GSM2433112	SAMN06165905	SRX2436142	IPF	47.84	9.66
SRR5120917	GSM2433113	SAMN06165904	SRX2436143	IPF	62.52	12.63
SRR5120918	GSM2433114	SAMN06165903	SRX2436144	IPF	45.27	9.14
SRR5120919	GSM2433115	SAMN06165902	SRX2436145	IPF	47.96	9.69
SRR5120920	GSM2433116	SAMN06165901	SRX2436146	IPF	44.18	8.92
SRR5120921	GSM2433117	SAMN06165900	SRX2436147	IPF	53.41	10.79
SRR5120922	GSM2433118	SAMN06165899	SRX2436148	Control	27.49	5.55
SRR5120923	GSM2433119	SAMN06165898	SRX2436149	Control	53.33	10.77
SRR5120924	GSM2433120	SAMN06165897	SRX2436150	Control	52.40	10.59
SRR5120925	GSM2433121	SAMN06165948	SRX2436151	Control	47.64	9.62
SRR5120926	GSM2433122	SAMN06165947	SRX2436152	Control	68.34	13.80
SRR5120927	GSM2433123	SAMN06165946	SRX2436153	Control	41.63	8.41
SRR5120928	GSM2433124	SAMN06165945	SRX2436154	Control	48.22	9.74
SRR5120929	GSM2433125	SAMN06165944	SRX2436155	Control	43.93	8.87
SRR5120930	GSM2433126	SAMN06165943	SRX2436156	Control	52.89	10.68
SRR5120931	GSM2433127	SAMN06165942	SRX2436157	Control	44.99	9.09
SRR5120932	GSM2433128	SAMN06165941	SRX2436158	Control	41.96	8.48
SRR5120933	GSM2433129	SAMN06165940	SRX2436159	Control	53.63	10.83
SRR5120934	GSM2433130	SAMN06165939	SRX2436160	Control	48.91	9.88
SRR5120935	GSM2433131	SAMN06165938	SRX2436161	Control	37.29	7.53
SRR5120936	GSM2433132	SAMN06165937	SRX2436162	Control	52.69	10.64
SRR5120937	GSM2433133	SAMN06165936	SRX2436163	Control	49.75	10.05
SRR5120938	GSM2433134	SAMN06165935	SRX2436164	Control	51.68	10.44
SRR5120939	GSM2433135	SAMN06165934	SRX2436165	Control	53.05	10.72
SRR5120940	GSM2433136	SAMN06165933	SRX2436166	Control	46.95	9.48

S3.2 Per-sample metrics

Tables S3a to S3d give the per-sample values summarized in Table III of the main article. Read-level values come from the MultiQC reports of the Pre-trim QC and Post-trim QC modules. Trimming and alignment values come from the MultiQC report of the full pipeline, alignment processing values from the SAMtools flagstat table and quantification values from the HTSeq-

count statistics exported by the platform. Mean quality is the mean of the FastQC quality scores along the read, averaged over the two files of each sample.

Table S3a. Raw read quality and trimming outcome per sample. GC content and duplication refer to the forward reads. Adapter is the highest adapter content at any position in either file. Trimming outcomes are shares of input read pairs.

Run	Group	Read pairs (M)	GC (%)	Duplication (%)	Mean quality	Adapter (%)	Both mates kept (%)	Forward only (%)	Reverse only (%)	Dropped (%)	Pairs kept (M)
SRR5120902	IPF	60.16	50	19.9	35.1	0.53	88.8	6.1	2.3	2.8	53.42
SRR5120903	IPF	53.19	50	16.8	35.0	0.68	88.0	6.6	2.4	2.9	46.84
SRR5120904	IPF	33.10	50	13.4	35.2	0.48	88.9	6.1	2.3	2.7	29.43
SRR5120905	IPF	49.80	49	16.5	34.9	0.43	87.9	6.3	2.5	3.3	43.77
SRR5120906	IPF	60.36	50	18.2	34.9	0.46	88.2	6.2	2.5	3.1	53.22
SRR5120907	IPF	34.69	50	14.7	35.0	0.46	88.6	6.0	2.5	2.8	30.75
SRR5120908	IPF	42.94	49	16.6	34.7	0.47	86.7	6.8	2.8	3.6	37.24
SRR5120909	IPF	45.67	48	15.6	34.9	0.68	87.0	6.6	3.1	3.3	39.72
SRR5120910	IPF	54.46	50	16.5	34.9	0.56	87.8	6.6	2.6	3.1	47.80
SRR5120911	IPF	57.04	53	12.1	34.5	0.33	86.9	6.8	2.8	3.6	49.55
SRR5120912	IPF	48.15	51	15.7	34.9	0.47	87.8	6.4	2.6	3.2	42.28
SRR5120913	IPF	39.52	50	14.0	34.7	0.38	87.1	6.8	2.6	3.5	34.43
SRR5120914	IPF	43.91	49	14.5	35.3	0.58	89.2	6.0	2.3	2.5	39.15
SRR5120915	IPF	42.69	48	16.1	35.3	0.63	89.3	5.6	2.4	2.6	38.14
SRR5120916	IPF	47.84	50	15.9	35.0	0.64	88.0	6.4	2.6	3.0	42.08
SRR5120917	IPF	62.52	51	17.8	34.7	0.43	87.0	6.8	2.9	3.3	54.38
SRR5120918	IPF	45.27	50	16.0	34.7	0.72	86.1	7.2	3.0	3.7	38.97
SRR5120919	IPF	47.96	51	14.9	34.9	0.45	88.2	6.2	2.6	3.0	42.29
SRR5120920	IPF	44.18	50	13.6	34.8	0.45	87.7	6.5	2.7	3.2	38.73
SRR5120921	IPF	53.41	51	17.8	35.0	0.66	88.0	6.5	2.6	2.9	47.02
SRR5120922	Control	27.49	46	16.6	36.0	1.15	90.5	5.7	2.1	1.7	24.89
SRR5120923	Control	53.33	45	21.7	36.3	1.10	92.0	4.5	2.1	1.4	49.09
SRR5120924	Control	52.40	46	16.3	36.2	0.72	92.5	4.1	1.8	1.6	48.46
SRR5120925	Control	47.64	43	21.9	36.5	1.55	91.2	5.6	2.0	1.3	43.43
SRR5120926	Control	68.34	48	18.9	35.9	0.55	91.8	4.3	1.9	1.9	62.76
SRR5120927	Control	41.63	48	16.2	36.4	0.63	93.0	4.2	1.4	1.4	38.71
SRR5120928	Control	48.22	49	17.2	35.8	0.49	91.4	4.7	1.8	2.1	44.08
SRR5120929	Control	43.93	49	17.9	35.7	0.69	90.7	5.2	2.0	2.1	39.84
SRR5120930	Control	52.89	50	17.7	36.0	0.82	91.9	4.9	1.4	1.7	48.62
SRR5120931	Control	44.99	49	16.7	35.7	0.60	91.2	5.0	1.8	2.0	41.04
SRR5120932	Control	41.96	44	27.4	36.6	2.37	90.3	6.9	1.8	1.1	37.87
SRR5120933	Control	53.63	49	17.1	35.0	0.46	87.9	6.3	2.6	3.2	47.16
SRR5120934	Control	48.91	47	17.6	36.4	0.73	92.9	4.3	1.5	1.3	45.43
SRR5120935	Control	37.29	49	14.5	35.7	0.46	91.0	5.1	1.8	2.1	33.93
SRR5120936	Control	52.69	48	16.5	35.9	0.77	91.2	4.8	2.0	2.0	48.07
SRR5120937	Control	49.75	48	17.9	36.0	0.75	91.8	4.5	1.9	1.9	45.66
SRR5120938	Control	51.68	48	17.1	35.8	0.59	91.3	4.7	1.9	2.1	47.16
SRR5120939	Control	53.05	49	17.8	35.7	0.48	91.1	4.9	1.8	2.2	48.34
SRR5120940	Control	46.95	49	16.3	36.0	0.53	92.0	4.2	1.9	1.9	43.20

Table S3b. Quality of the trimmed reads per sample, from the MultiQC report of the Post-trim QC module. GC content refers to the forward reads, and adapter is the highest adapter content at any position in either file.

Run	Group	Sequences per file (M)	Mean length R1 (bp)	Mean length R2 (bp)	Full 101 bp, R1 (%)	GC (%)	Duplication R1 (%)	Duplication R2 (%)	Mean quality	Adapter (%)
SRR5120902	IPF	53.42	98.7	98.7	88.4	50	19.7	19.8	36.6	0.21
SRR5120903	IPF	46.84	98.6	98.6	88.0	50	16.2	16.3	36.6	0.22
SRR5120904	IPF	29.43	98.7	98.7	88.3	49	12.9	12.8	36.6	0.24
SRR5120905	IPF	43.77	98.7	98.6	88.0	49	15.9	16.2	36.6	0.24
SRR5120906	IPF	53.22	98.5	98.7	87.2	49	18.2	18.1	36.5	0.26
SRR5120907	IPF	30.75	98.5	98.8	87.2	49	14.2	14.2	36.5	0.25
SRR5120908	IPF	37.24	98.5	98.5	87.4	48	16.0	16.1	36.5	0.26
SRR5120909	IPF	39.72	98.5	98.7	87.4	47	15.5	15.5	36.6	0.35
SRR5120910	IPF	47.80	98.4	98.7	86.6	49	16.5	16.6	36.5	0.23
SRR5120911	IPF	49.55	98.3	98.4	86.0	52	11.6	11.6	36.3	0.18
SRR5120912	IPF	42.28	98.4	98.6	86.6	50	15.1	15.1	36.5	0.19
SRR5120913	IPF	34.43	98.5	98.3	87.3	50	13.7	13.6	36.4	0.21
SRR5120914	IPF	39.15	98.7	98.8	88.5	48	13.7	13.6	36.7	0.27
SRR5120915	IPF	38.14	98.8	99.0	88.6	47	16.4	16.6	36.8	0.35
SRR5120916	IPF	42.08	98.6	98.7	87.8	49	15.3	15.3	36.6	0.23
SRR5120917	IPF	54.38	98.4	98.5	86.8	50	18.0	18.3	36.4	0.22
SRR5120918	IPF	38.97	98.4	98.5	87.0	49	15.2	15.4	36.4	0.22
SRR5120919	IPF	42.29	98.4	98.7	86.5	51	14.4	14.4	36.5	0.20
SRR5120920	IPF	38.73	98.3	98.6	86.4	50	12.9	12.7	36.5	0.24
SRR5120921	IPF	47.02	98.6	98.7	87.6	50	17.2	17.6	36.5	0.23
SRR5120922	Control	24.89	99.1	99.3	90.6	45	15.7	15.1	37.1	0.61
SRR5120923	Control	49.09	99.5	99.6	87.8	45	21.1	21.4	37.5	0.63
SRR5120924	Control	48.46	99.6	99.5	87.8	46	16.3	16.9	37.4	0.42
SRR5120925	Control	43.43	99.7	99.7	88.8	43	21.4	21.7	37.6	0.59
SRR5120926	Control	62.76	99.4	99.4	86.8	48	19.0	19.3	37.2	0.32
SRR5120927	Control	38.71	99.7	99.6	88.0	47	15.8	16.0	37.5	0.37
SRR5120928	Control	44.08	99.5	99.3	86.7	49	16.8	17.3	37.2	0.28
SRR5120929	Control	39.84	99.3	99.2	85.8	49	16.9	17.5	37.1	0.40
SRR5120930	Control	48.62	99.6	99.4	86.5	50	17.3	17.2	37.2	0.26
SRR5120931	Control	41.04	99.2	99.1	90.6	48	16.2	15.9	37.0	0.34
SRR5120932	Control	37.87	99.6	99.7	88.6	44	26.3	26.6	37.7	0.74
SRR5120933	Control	47.16	98.6	98.6	87.9	48	16.6	16.8	36.6	0.25
SRR5120934	Control	45.43	99.7	99.6	88.0	47	18.0	18.1	37.5	0.42
SRR5120935	Control	33.93	99.2	99.1	90.4	49	14.0	13.7	36.9	0.26
SRR5120936	Control	48.07	99.4	99.2	86.4	48	15.9	16.1	37.2	0.45
SRR5120937	Control	45.66	99.5	99.4	87.0	48	18.7	19.2	37.3	0.44
SRR5120938	Control	47.16	99.4	99.3	86.6	48	16.1	16.7	37.2	0.34
SRR5120939	Control	48.34	99.5	99.2	86.2	49	17.9	18.2	37.1	0.28
SRR5120940	Control	43.20	99.5	99.4	86.8	48	16.6	16.9	37.3	0.31

Table S3c. STAR alignment metrics per sample. Shares are of input read pairs. Splices are the number of spliced reads reported by STAR, and mapped length is the mean length of the aligned pair in bases.

Run	Group	Input pairs (M)	Aligned (%)	Unique (%)	Multiple loci (%)	Too many loci (%)	Too short (%)	Mismatch rate (%)	Splices (M)	Annotated splices (%)	Mapped length
SRR5120902	IPF	53.42	97.9	70.2	27.7	0.03	1.7	0.17	21.1	98.6	197.3
SRR5120903	IPF	46.84	98.0	66.2	31.8	0.04	1.6	0.15	17.1	98.6	197.3
SRR5120904	IPF	29.43	98.3	71.9	26.4	0.04	1.5	0.14	10.8	98.8	197.4
SRR5120905	IPF	43.77	97.7	72.4	25.3	0.04	1.7	0.15	16.9	98.4	197.2
SRR5120906	IPF	53.22	98.0	75.3	22.7	0.04	1.6	0.15	22.6	98.7	197.1
SRR5120907	IPF	30.75	97.8	73.8	24.0	0.04	1.7	0.15	12.1	98.6	197.2
SRR5120908	IPF	37.24	98.7	80.6	18.1	0.03	1.1	0.15	16.7	99.0	196.9
SRR5120909	IPF	39.72	98.7	81.5	17.2	0.04	1.1	0.14	14.1	98.8	197.1
SRR5120910	IPF	47.80	98.7	70.3	28.4	0.04	1.2	0.13	18.5	99.0	197.2
SRR5120911	IPF	49.55	96.1	47.0	49.0	0.04	3.1	0.21	11.1	96.4	196.8
SRR5120912	IPF	42.28	97.7	65.3	32.4	0.03	1.8	0.14	16.0	98.7	197.2
SRR5120913	IPF	34.43	98.6	63.5	35.0	0.04	1.2	0.14	12.2	98.8	197.2
SRR5120914	IPF	39.15	98.1	72.7	25.3	0.04	1.5	0.15	13.8	98.5	197.6
SRR5120915	IPF	38.14	98.3	81.0	17.3	0.04	1.4	0.15	12.7	98.4	197.7
SRR5120916	IPF	42.08	98.1	66.8	31.3	0.04	1.6	0.14	15.1	98.7	197.6
SRR5120917	IPF	54.38	96.7	68.5	28.2	0.03	2.7	0.18	21.4	97.8	196.7
SRR5120918	IPF	38.97	97.8	70.3	27.5	0.03	1.6	0.15	15.9	98.6	197.0
SRR5120919	IPF	42.29	96.8	62.8	33.9	0.04	2.5	0.19	14.5	98.1	197.1
SRR5120920	IPF	38.73	98.3	62.6	35.7	0.04	1.4	0.15	11.8	98.6	197.2
SRR5120921	IPF	47.02	97.4	66.3	31.1	0.03	2.1	0.17	17.5	98.5	197.3
SRR5120922	Control	24.89	98.3	85.6	12.7	0.08	1.4	0.15	5.4	98.3	198.2
SRR5120923	Control	49.09	98.6	87.3	11.3	0.06	1.1	0.14	14.0	98.5	198.8
SRR5120924	Control	48.46	98.6	85.9	12.7	0.04	1.2	0.13	18.1	98.7	198.8
SRR5120925	Control	43.43	98.5	86.8	11.8	0.05	1.2	0.14	13.6	98.8	198.9
SRR5120926	Control	62.76	98.5	82.4	16.1	0.04	1.2	0.14	25.0	98.8	198.6
SRR5120927	Control	38.71	98.4	83.8	14.7	0.04	1.3	0.13	15.0	98.7	199.0
SRR5120928	Control	44.08	98.5	77.2	21.3	0.04	1.3	0.13	18.0	98.8	198.5
SRR5120929	Control	39.84	98.4	81.0	17.4	0.05	1.4	0.15	13.7	98.7	198.3
SRR5120930	Control	48.62	98.3	67.4	30.9	0.04	1.4	0.15	16.9	98.7	198.8
SRR5120931	Control	41.04	98.3	82.1	16.1	0.05	1.5	0.13	15.4	98.7	198.0
SRR5120932	Control	37.87	98.4	87.4	11.0	0.06	1.3	0.13	10.0	98.6	198.8
SRR5120933	Control	47.16	98.8	76.7	22.1	0.03	1.1	0.14	19.3	98.9	197.3
SRR5120934	Control	45.43	98.6	83.4	15.2	0.04	1.2	0.13	15.7	98.8	199.0
SRR5120935	Control	33.93	98.5	75.6	22.9	0.04	1.4	0.13	13.4	98.8	198.1
SRR5120936	Control	48.07	98.2	83.0	15.3	0.06	1.6	0.14	15.2	98.5	198.4
SRR5120937	Control	45.66	98.5	86.0	12.5	0.04	1.3	0.14	15.9	98.7	198.5
SRR5120938	Control	47.16	98.3	83.1	15.2	0.04	1.5	0.14	18.9	98.9	198.4
SRR5120939	Control	48.34	98.6	70.6	28.0	0.04	1.3	0.15	17.3	98.9	198.6
SRR5120940	Control	43.20	97.9	80.7	17.2	0.04	1.7	0.14	18.2	98.6	198.6

Table S3d. Alignment processing and quantification per sample. Primary, primary alignments. Dup., duplicates as a share of primary alignments. Proper, properly paired alignments. Counted, read pairs counted by HTSeq-count. Rate, assignment rate, which is assigned pairs divided by the sum of assigned, no_feature and ambiguous pairs. No feature, ambiguous (Ambig.) and not unique pairs are shares of all counted pairs. Features, annotation features with at least one assigned read.

Run	Group	Primary (M)	Dup. (%)	Proper (%)	Singletons	Counted (M)	Assigned (M)	Rate (%)	No feature (%)	Ambig. (%)	Not unique (%)	Features
SRR5120902	IPF	105.1	36.5	99.999	986	52.6	10.17	27.0	4.5	47.8	28.4	60,728
SRR5120903	IPF	92.2	36.7	99.999	1,071	46.1	8.45	27.2	5.5	43.6	32.6	60,195
SRR5120904	IPF	58.0	28.1	99.999	592	29.0	5.69	26.8	7.4	46.0	26.9	57,352
SRR5120905	IPF	86.2	30.9	99.999	1,014	43.1	8.43	26.5	6.7	47.7	26.0	61,470
SRR5120906	IPF	105.0	30.9	99.999	1,084	52.5	10.59	26.3	6.0	50.5	23.3	64,627
SRR5120907	IPF	60.5	26.7	99.999	640	30.3	6.01	26.4	6.3	49.2	24.7	55,736
SRR5120908	IPF	73.7	23.4	99.999	797	36.8	7.54	25.1	6.1	55.1	18.3	58,517
SRR5120909	IPF	78.4	25.9	99.999	1,042	39.2	8.60	26.6	11.6	49.0	17.4	67,280
SRR5120910	IPF	94.4	33.3	99.999	891	47.2	8.56	25.4	7.1	46.0	28.8	64,401
SRR5120911	IPF	96.7	52.1	99.999	1,019	48.3	7.12	30.1	5.8	28.5	51.0	62,304
SRR5120912	IPF	83.1	35.8	99.999	733	41.5	7.38	26.7	5.3	43.6	33.3	57,427
SRR5120913	IPF	68.1	37.4	99.999	670	34.0	5.66	25.8	6.0	41.9	35.5	56,541
SRR5120914	IPF	77.1	30.8	99.999	843	38.6	7.58	26.6	9.3	45.1	26.0	65,548
SRR5120915	IPF	75.2	27.2	99.999	850	37.6	8.94	28.9	10.3	48.3	17.6	66,838
SRR5120916	IPF	82.8	34.5	99.999	817	41.4	7.53	26.7	6.5	43.4	32.0	61,623
SRR5120917	IPF	106.3	36.3	99.999	1,267	53.2	10.70	28.5	3.6	46.9	29.4	57,286
SRR5120918	IPF	76.6	32.8	99.999	896	38.3	7.33	26.7	5.0	47.5	28.3	56,972
SRR5120919	IPF	82.7	37.9	99.999	803	41.3	7.23	27.0	5.7	41.5	35.2	60,050
SRR5120920	IPF	76.4	37.3	99.999	785	38.2	6.25	25.7	8.9	38.5	36.3	64,079
SRR5120921	IPF	92.2	35.8	99.999	1,023	46.1	8.17	26.1	5.2	45.0	32.1	59,496
SRR5120922	Control	48.8	28.3	99.999	685	24.4	5.96	28.1	20.9	41.7	12.9	63,331
SRR5120923	Control	96.8	33.1	99.999	1,192	48.4	12.33	28.8	14.6	48.5	11.4	69,403
SRR5120924	Control	95.7	26.9	99.999	801	47.8	10.81	25.9	13.3	51.3	12.8	72,329
SRR5120925	Control	84.9	39.8	99.999	755	42.4	11.21	30.0	12.3	49.3	11.9	64,201
SRR5120926	Control	124.1	31.3	99.999	1,130	62.0	14.07	27.1	9.6	51.3	16.4	70,388
SRR5120927	Control	76.2	25.0	99.999	703	38.1	8.74	27.0	10.3	51.9	14.9	66,293
SRR5120928	Control	87.1	27.6	99.999	697	43.5	9.10	26.6	7.3	50.2	21.6	62,114
SRR5120929	Control	78.5	28.7	99.999	948	39.3	9.01	27.9	10.1	49.3	17.6	63,571
SRR5120930	Control	95.8	36.8	99.999	750	47.9	8.58	26.1	7.3	43.3	31.5	63,506
SRR5120931	Control	81.0	23.1	99.999	1,070	40.5	8.95	26.4	10.2	51.3	16.4	66,823
SRR5120932	Control	73.5	46.3	99.999	965	36.8	10.76	33.0	15.8	43.7	11.2	63,362
SRR5120933	Control	93.3	27.7	99.999	1,017	46.7	9.48	26.2	7.4	50.0	22.3	63,840
SRR5120934	Control	89.5	29.2	99.999	837	44.8	10.09	26.6	11.8	50.3	15.4	67,888
SRR5120935	Control	67.0	26.3	99.999	720	33.5	6.76	26.3	8.0	48.6	23.2	59,756
SRR5120936	Control	94.6	22.0	99.999	1,209	47.3	10.12	25.3	14.6	48.6	15.5	74,216
SRR5120937	Control	90.2	26.1	99.999	1,163	45.1	10.88	27.6	9.1	54.1	12.7	62,664
SRR5120938	Control	93.0	22.4	99.999	908	46.5	10.36	26.3	9.5	52.8	15.4	69,505
SRR5120939	Control	95.5	33.7	99.999	819	47.8	9.00	26.3	7.5	45.3	28.4	63,530
SRR5120940	Control	85.1	25.5	99.999	751	42.5	9.12	26.0	9.3	51.7	17.6	66,415

S3.3 Module settings

Table S4 lists the settings of every module as they were read back from the job records of the case study. Settings that are not listed were left at the defaults of the engine concerned.

Table S4. Settings used in the case study, by module.

Module	Engine	Setting	Value
1 Upload	fasterq-dump	Input	39 SRA run accessions, SRR5120902 to SRR5120940
		Output	Forward and reverse FASTQ files for each run, with integrity and pairing checks
2 Pre-trim QC	FastQC, MultiQC	Mode	Each of the 78 files analyzed separately with default settings, then aggregated
3 Trimming	Trimmomatic	Mode	Paired-end, Phred+33 quality encoding, 4 threads
		Adapter file	TruSeq2-PE.fa
		ILLUMINACLIP	2:30:10
		LEADING, TRAILING	3, 3
		SLIDINGWINDOW	4:15
		MINLEN	50
		HEADCROP, TAILCROP	Not applied
		Output used	Pairs in which both mates survived
4 Post-trim QC	FastQC, MultiQC	Mode	As for module 2, on the trimmed paired files
5 Alignment	STAR	Reference	UCSC hg38 (hg38.fa), with hg38.knownGene.gtf supplied through --sjdbGTFfile when the index was built
		Input	Trimmed forward and reverse reads of each sample
		Output	--outSAMtype BAM SortedByCoordinate, --outSAMattributes NH HI NM MD AS XS, --outSAMstrandField intronMotif, --quantMode GeneCounts
		Junctions	--alignSJDBoverhangMin 10
6 Pre-quantification	SAMtools	Steps	sort, fixmate, markdup (duplicates flagged and kept), index, flagstat
7 Quantification	HTSeq-count	Options	-f bam, -r pos, -t exon, -i gene_id, -s no
		Output	Per-sample counts merged into one feature-by-sample matrix
8 Annotation	g:Profiler g:Convert	Target	Ensembl gene identifiers, HGNC symbols and descriptions for Homo sapiens
		Unresolved identifiers	Removed from the annotated matrix
9 DEG analysis	PyDESeq2	Design	~Condition, IPF (20 samples) and control (19 samples)
		Model	Median-of-ratios size factors, dispersion trend fitting and shrinkage, Wald test, Benjamini and Hochberg adjustment
		Filter	Features with baseMean below 10 excluded before calling
		Significance	Adjusted P below 0.05 and absolute $\log_2$ fold change above 0.5
		Contrast	Control against IPF, with control as the numerator, so negative $\log_2$ fold changes denote features raised in IPF
	GSEApy with Enrichr	Over-representation analysis	100 unique symbols with the smallest adjusted P among the differentially expressed features, GO_Biological_Process_2023, default Enrichr background
	GSEApy	Preranked GSEA	All tested genes ranked by $\log_2$ fold change, GO_Biological_Process_2023, 100 permutations
10 Gene Splicing	STAR	Threads	30
		Chimeric settings	--chimSegmentMin 12, --chimOutType WithinBAM SoftClip, --chimJunctionOverhangMin 8, --chimOutJunctionFormat 1, --chimMultimapNmax 20, --chimMultimapScoreRange 3, --chimNonchimScoreDropMin 10, --chimScoreJunctionNonGTAG -4
		Junction and pairing settings	--alignSJDBoverhangMin 10, --alignMatesGapMax 100000, --alignIntronMax 100000, --alignSJstitchMismatchNmax 5 -1 5 5, --peOverlapNbasesMin 12, --peOverlapMMp 0.1
		Output	Coordinate-sorted BAM with NH HI NM MD AS XS attributes and --outSAMstrandField intronMotif
	splicekit with edgeR	Contrast and threshold	IPF versus control, FDR below 0.05

Module	Engine	Setting	Value
11 Gene fusion Detection	STAR	Settings recommended for Arriba	--outFilterMultimapNmax 50, --peOverlapNbasesMin 10, --alignSplicedMateMapLminOverLmate 0.5, --alignSJstitchMismatchNmax 5 -1 5 5, --chimSegmentMin 10, --chimOutType WithinBAM HardClip, --chimJunctionOverhangMin 10, --chimScoreDropMax 30, --chimScoreJunctionNonGTAG 0, --chimScoreSeparation 1, --chimSegmentReadGapMax 3, --chimMultimapNmax 50
	Arriba	Inputs	STAR BAM file, genome FASTA and annotation matching the STAR index, blacklist, known fusions and protein domain files of release 2.5.0
		Outputs	Fusion table, discarded candidates and one PDF of visualizations per sample

S3.4 Differential expression

Tables S5a and S5b extend Table IV of the main article to the 50 most significant up-regulated and down-regulated features, from the 1,714 and 1,750 features in the two sets. Fold changes are expressed as control relative to IPF, as exported by the DEG analysis module, so the down-regulated features are those raised in IPF. Symbols are those assigned by the platform. The complete result for all 10,484 tested features is given in Supplementary Data S1.

Table S5a. The 50 most significant up-regulated features, ranked by adjusted P. Fold changes are expressed as control relative to IPF, so these features are reduced in IPF. Features are identified by their UniProt accessions in the count matrix.

Rank	Feature	Symbol	baseMean	$\log_2$ fold change	Standard error	Adjusted P
1	Q9HCM4	*EPB41L5*	1,430	1.71	0.12	4.3×10^{-43}
2	P46721	*SLCO1A2*	134	6.55	0.49	2.5×10^{-36}
3	Q8IUC8	*GALNT13*	57	3.86	0.30	2.1×10^{-33}
4	P54851	*EMP2*	17,804	1.94	0.15	9.1×10^{-33}
5	Q16825	*PTPN21*	954	1.51	0.13	2.4×10^{-28}
6	Q717R9	*CYS1*	254	2.14	0.18	3.0×10^{-28}
7	Q9NUG4	*CCM2L*	264	1.77	0.16	4.2×10^{-27}
8	Q8WXH6	*RAB40A*	91	2.27	0.20	4.3×10^{-27}
9	Q96KS9	*FAM167A*	356	2.13	0.19	1.5×10^{-26}
10	Q8N0Z9	*VSIG10*	610	1.67	0.15	2.9×10^{-26}
11	Q12851	*MAP4K2*	253	1.86	0.17	9.4×10^{-26}
12	Q6FH91	*TNNC1*	87	2.77	0.25	4.0×10^{-25}
13	Q8N987	*NECAB1*	202	1.81	0.17	3.3×10^{-24}
14	Q6ZS82	*RGS9BP*	16	2.75	0.26	1.6×10^{-23}
15	Q8IZC4	*RTKN2*	4,605	4.01	0.38	2.6×10^{-23}
16	Q9Y2G0	*EFR3B*	172	1.77	0.17	4.7×10^{-23}
17	V9GYT2	*ANKRD29*	107	2.06	0.20	2.1×10^{-22}
18	X5DQM5	*ADRB2*	747	1.35	0.13	2.1×10^{-22}
19	P51826	*AFF3*	80	2.44	0.24	2.4×10^{-22}
20	H0YMK1	*DAPK2*	150	1.93	0.19	2.5×10^{-22}
21	Q53X57	*CAV2*	4,511	1.59	0.15	3.2×10^{-22}
22	O75113	*N4BP1*	1,007	0.84	0.08	3.7×10^{-22}
23	P50225	*SULT1A1*	29	2.11	0.21	6.1×10^{-22}
24	Q5W111	*SPRYD7*	632	1.09	0.11	6.1×10^{-22}
25	Q8N9R8	*SCAI*	696	1.20	0.12	9.0×10^{-22}
26	Q6UXG8	*BTNL9*	2,398	3.00	0.29	1.1×10^{-21}

Rank	Feature	Symbol	baseMean	$\log_2$ fold change	Standard error	Adjusted P
27	Q9BT76	*UPK3B*	198	3.69	0.36	1.1×10^{-21}
28	Q8IV16	*GPIHBP1*	1,356	2.57	0.25	1.3×10^{-21}
29	Q8N6Y0	*USHBP1*	73	2.25	0.22	2.0×10^{-21}
30	A4D1S0	*KLRG2*	75	2.60	0.26	2.6×10^{-21}
31	Q8N6D5	*ANKRD29*	378	2.20	0.22	2.7×10^{-21}
32	Q86T90	*KIAA1328*	119	0.81	0.08	8.1×10^{-21}
33	Q5SVY4	*RTKN2*	259	4.59	0.46	9.2×10^{-21}
34	A2A2V5	*SERTM1*	181	4.15	0.42	2.0×10^{-20}
35	Q9H2E6	*SEMA6A*	594	1.85	0.19	3.6×10^{-20}
36	A0A1B0GVN0	*TINCR*	89	2.19	0.22	3.6×10^{-20}
37	Q9Y5W3	*KLF2*	998	1.97	0.20	5.1×10^{-20}
38	P43005	*SLC1A1*	922	1.55	0.16	5.2×10^{-20}
39	Q15172	*PPP2R5A*	2,936	0.67	0.07	5.2×10^{-20}
40	Q9HDC5	*JPH1*	237	1.42	0.15	7.1×10^{-20}
41	P35590	*TIE1*	519	1.30	0.13	8.3×10^{-20}
42	P51674	*GPM6A*	96	2.54	0.26	2.1×10^{-19}
43	Q9HB89	*NMUR1*	133	2.19	0.23	2.8×10^{-19}
44	P33764	*PRH1*	199	2.33	0.24	3.7×10^{-19}
45	Q5TCX8	*MAP3K21*	291	1.27	0.13	6.5×10^{-19}
46	Q9UKF5	*ADAM29*	11	3.93	0.42	1.7×10^{-18}
47	Q9NUR3	*TMEM74B*	190	1.90	0.20	2.5×10^{-18}
48	Q9Y342	*PLLP*	603	1.81	0.20	3.3×10^{-18}
49	Q9BYJ0	*FGFBP2*	50	2.77	0.30	4.3×10^{-18}
50	Q9Y287	*ITM2B*	114	1.29	0.14	1.0×10^{-17}

Table S5b. The 50 most significant down-regulated features, ranked by adjusted P. Fold changes are expressed as control relative to IPF, so these features are raised in IPF. Features are identified by their UniProt accessions in the count matrix.

Rank	Feature	Symbol	baseMean	$\log_2$ fold change	Standard error	Adjusted P
1	Q9UMD9	*COL17A1*	166	−4.79	0.32	3.3×10^{-47}
2	P09238	*MMP10*	191	−6.19	0.48	4.7×10^{-35}
3	P03956	*MMP1*	553	−7.67	0.59	1.2×10^{-34}
4	Q8N0V4	*LGI2*	235	−2.87	0.23	1.0×10^{-32}
5	Q9HCM2	*PLXNA4*	724	−3.20	0.26	1.2×10^{-31}
6	P54886	*ALDH18A1*	620	−1.09	0.09	1.8×10^{-29}
7	Q86UK0	*ABCA12*	46	−3.88	0.33	3.9×10^{-29}
8	Q9H6B4	*CLMP*	1,154	−2.03	0.17	7.1×10^{-29}
9	Q03692	*COL10A1*	286	−3.78	0.32	1.2×10^{-28}
10	P36269	*GGT5*	181	−2.24	0.19	6.1×10^{-28}
11	P02461	*COL3A1*	25,533	−3.59	0.32	6.6×10^{-27}
12	P45452	*MMP13*	93	−5.62	0.50	8.1×10^{-27}
13	P29323	*L3MBTL2-AS1*	149	−1.94	0.17	1.0×10^{-26}
14	P21217	*FUT3*	135	−1.70	0.15	2.9×10^{-26}
15	E7EW31	*PROB1*	93	−1.49	0.13	5.5×10^{-26}
16	P22223	*CDH3*	106	−2.60	0.23	9.4×10^{-26}
17	P08138	*NGFR*	184	−2.82	0.25	9.4×10^{-26}
18	Q8IZW8	*TNS4*	128	−4.07	0.37	9.8×10^{-26}
19	Q86UY5	*FAM83A*	114	−4.35	0.40	8.5×10^{-25}
20	Q99727	*TIMP4*	67	−3.05	0.28	1.3×10^{-24}
21	Q9Y617	*PSAT1*	268	−3.02	0.28	1.4×10^{-24}
22	P09237	*MMP7*	1,362	−3.81	0.35	1.9×10^{-24}

Rank	Feature	Symbol	baseMean	$\log_2$ fold change	Standard error	Adjusted P
23	O43895	*XPNPEP2*	94	−2.93	0.27	2.4×10^{-24}
24	Q9HCB6	*SPON1*	3,888	−1.70	0.16	1.5×10^{-23}
25	O15347	*HMGB3*	768	−1.93	0.18	3.1×10^{-23}
26	P05154	*SERPINA5*	31	−3.17	0.30	3.4×10^{-23}
27	Q03001	*DST*	589	−2.57	0.24	5.0×10^{-23}
28	P29120	*PCSK1*	52	−3.73	0.36	6.1×10^{-23}
29	Q9P2B2	*PTGFRN*	1,461	−2.12	0.20	6.1×10^{-23}
30	Q14627	*IL13RA2*	170	−3.83	0.37	7.8×10^{-23}
31	Q8IV01	*SYT12*	33	−3.44	0.33	1.3×10^{-22}
32	Q96CG8	*CTHRC1*	138	−3.19	0.31	1.6×10^{-22}
33	O60565	*GREM1*	54	−6.66	0.64	2.1×10^{-22}
34	Q05707	*COL14A1*	4,524	−2.47	0.24	3.1×10^{-22}
35	O15484	*CAPN5*	198	−1.45	0.14	3.8×10^{-22}
36	Q13641	*TPBG*	552	−1.30	0.13	1.2×10^{-21}
37	Q8N271	*PROM2*	52	−2.72	0.27	1.4×10^{-21}
38	B1AKI9	*ISM1*	482	−1.98	0.20	2.3×10^{-21}
39	O75509	*TNFRSF21*	1,922	−1.44	0.14	6.8×10^{-21}
40	Q02388	*COL7A1*	220	−2.24	0.23	1.3×10^{-20}
41	Q8WYA1	*BMAL2*	338	−1.99	0.20	1.6×10^{-20}
42	Q9H9B4	*SFXN1*	444	−0.66	0.07	2.1×10^{-20}
43	P02538	*KRT6A*	458	−7.45	0.76	2.1×10^{-20}
44	P47972	*NPTX2*	66	−3.08	0.31	2.5×10^{-20}
45	P02452	*COL1A1*	9,997	−3.61	0.37	2.6×10^{-20}
46	Q9NYQ8	*FAT2*	139	−3.91	0.40	3.1×10^{-20}
47	O75339	*CILP*	367	−2.83	0.29	3.2×10^{-20}
48	Q8WXS8	*ADAMTS14*	269	−2.90	0.30	4.4×10^{-20}
49	O60287	*URB1*	1,001	−1.41	0.15	5.1×10^{-20}
50	Q10981	*FUT2*	209	−2.81	0.29	5.1×10^{-20}

S3.5 Functional enrichment

Table S6a lists the 20 GO Biological Process terms with the smallest adjusted P values in the over-representation analysis of the 100 most significant genes. Enrichr returned 753 terms in total, and all of them are given in Supplementary Data S1. Table S6b lists the 17 gene sets with an FDR below 0.05 in the preranked GSEA. Genes were ranked by the $\log_2$ fold change of control relative to IPF, so a negative normalized enrichment score denotes enrichment among the down-regulated genes, which are raised in IPF. Nominal P values of zero, which arise with 100 permutations, are shown as below 0.01.

Table S6a. Over-representation analysis of the 100 most significant differentially expressed genes against GO Biological Process 2023 through Enrichr. Overlap is the number of input genes in the term over the size of the term.

Term	GO identifier	Overlap	P	Adjusted P	Odds ratio	Combined score	Genes
Extracellular Structure Organization	GO:0043062	12/109	1.8×10^{-13}	7.5×10^{-11}	28.8	846	*COL17A1, COL1A1, ADAMTS14, COL3A1, COL1A2, MMP7, MMP13, COL14A1, MMP1, COL7A1, COL10A1, MMP10*
External Encapsulating Structure Organization	GO:0045229	12/110	2.0×10^{-13}	7.5×10^{-11}	28.5	834	*COL17A1, COL1A1, ADAMTS14, COL3A1, COL1A2, MMP7, MMP13, COL14A1, MMP1, COL7A1, COL10A1, MMP10*
Extracellular Matrix Organization	GO:0030198	13/176	3.0×10^{-12}	7.5×10^{-10}	18.7	497	*COL17A1, MMP7, COL14A1, MMP1, MMP10, COL1A1, GREM1, ADAMTS14, COL3A1, COL1A2, MMP13, COL7A1, COL10A1*
Collagen Fibril Organization	GO:0030199	5/42	1.8×10^{-6}	3.4×10^{-4}	29.2	386	*COL1A1, GREM1, COL3A1, COL1A2, COL14A1*
Extracellular Matrix Disassembly	GO:0022617	4/44	6.1×10^{-5}	0.009	21.4	207	*MMP7, MMP13, MMP1, MMP10*
Cellular Component Disassembly	GO:0022411	4/46	7.3×10^{-5}	0.009	20.3	194	*MMP7, MMP13, MMP1, MMP10*
Supramolecular Fiber Organization	GO:0097435	8/316	1.5×10^{-4}	0.016	5.7	50	*COL1A1, GREM1, ADAMTS14, COL3A1, COL1A2, COL14A1, EMP2, KRT6A*
Plasma Membrane Raft Assembly	GO:0044854	2/5	2.3×10^{-4}	0.022	139.6	1170	*CAV2, EMP2*
Regulation Of Hydrolase Activity	GO:0051336	4/71	3.9×10^{-4}	0.033	12.7	100	*CAV1, PLXNA4, SERPINA5, PROM2*
Regulation Of Pinocytosis	GO:0048548	2/8	6.4×10^{-4}	0.044	69.8	513	*CAV1, PROM2*
Skin Morphogenesis	GO:0043589	2/8	6.4×10^{-4}	0.044	69.8	513	*COL1A1, COL1A2*
Negative Regulation Of Endothelial Cell Proliferation	GO:0001937	3/37	7.6×10^{-4}	0.048	18.7	134	*NGFR, CAV2, CAV1*
Negative Regulation Of Sprouting Angiogenesis	GO:1903671	2/10	0.0010	0.055	52.4	361	*SEMA6A, KLF2*
Receptor-Mediated Endocytosis Of Virus By Host Cell	GO:0019065	2/10	0.0010	0.055	52.4	361	*CAV2, CAV1*
Regulation Of Endothelial Cell Proliferation	GO:0001936	4/93	0.0011	0.055	9.6	65	*CAV2, CAV1, TIE1, FUT2*
Response To Amyloid-Beta	GO:1904645	3/43	0.0012	0.056	15.8	107	*NGFR, MMP13, ADRB2*
Regulation Of Intrinsic Apoptotic Signaling Pathway	GO:2001242	3/48	0.0016	0.072	14.1	90	*CAV1, DAPK2, RTKN2*
Negative Regulation Of Transmembrane Receptor Protein Serine/Threonine Kinase Signaling Pathway	GO:0090101	4/111	0.0021	0.088	8.0	49	*GREM1, CILP, CAV1, CAPN5*
Vasodilation	GO:0042311	2/15	0.0023	0.093	32.2	195	*ADRB2, KLF2*
Cellular Response To Peptide	GO:1901653	3/56	0.0025	0.096	12.0	71	*NGFR, CAV1, ADRB2*

Table S6b. Gene sets of GO Biological Process 2023 with an FDR below 0.05 in preranked GSEA of all tested genes. Negative NES denotes enrichment among down-regulated genes, which are raised in IPF. Tag gives the number of leading-edge genes over the number of set genes in the ranked list, and the first eight leading-edge genes are shown.

Gene set	GO identifier	NES	Nominal P	FDR	Tag	Leading-edge genes
External Encapsulating Structure Organization	GO:0045229	−2.30	< 0.01	< 0.001	42/84	*MMP1, MMP10, MMP13, COL17A1, MMP7, COL10A1, MMP8, COL3A1, ...*
Negative Regulation Of Endopeptidase Activity	GO:0010951	−2.29	< 0.01	< 0.001	12/44	*SERPINB4, CST1, SERPINB5, CST2, SERPINB2, SERPINA5, TIMP4, SERPINA11, ...*
Extracellular Matrix Organization	GO:0030198	−2.29	< 0.01	< 0.001	64/133	*MMP1, GREM1, PRSS2, MMP10, MMP13, COL17A1, MMP7, COL10A1, ...*
Extracellular Structure Organization	GO:0043062	−2.26	< 0.01	< 0.001	41/84	*MMP1, MMP10, MMP13, COL17A1, MMP7, COL10A1, MMP8, COL3A1, ...*
Cellular Component Disassembly	GO:0022411	−2.18	< 0.01	< 0.001	9/31	*MMP1, PRSS2, MMP10, MMP13, MMP7, MMP8, MMP3, MMP12, ...*
Regulation Of Endopeptidase Activity	GO:0052548	−2.14	< 0.01	< 0.001	9/45	*SERPINB4, SERPINB5, SERPINB2, SERPINA5, TIMP4, SERPINA11, SERPINB7, SERPINB13, ...*
Extracellular Matrix Disassembly	GO:0022617	−2.13	< 0.01	< 0.001	9/29	*MMP1, PRSS2, MMP10, MMP13, MMP7, MMP8, MMP3, MMP12, ...*
Intermediate Filament Organization	GO:0045109	−2.13	< 0.01	< 0.001	13/35	*KRT14, KRT6A, KRT75, KRT6B, KRT16, KRT5, DES, KRT17, ...*
Negative Regulation Of Peptidase Activity	GO:0010466	−2.05	< 0.01	0.001	9/42	*SERPINB4, SERPINB5, SERPINB2, SERPINA5, TIMP4, SERPINA11, SERPINB7, SERPINB13, ...*
Collagen Fibril Organization	GO:0030199	−2.07	< 0.01	0.002	18/35	*GREM1, COL3A1, COL1A2, CYP1B1, COL14A1, COL18A1, TLL2, ADAMTS2, ...*
Epidermis Development	GO:0008544	−2.02	< 0.01	0.002	23/62	*KRT14, SPRR1B, SPRR1A, COL17A1, KRT5, SPRR3, CALML5, GJB5, ...*
Cilium Movement	GO:0003341	−2.01	< 0.01	0.004	30/44	*CFAP206, HYDIN, CFAP53, RSPH4A, DNAH7, RSPH9, DNAAF6, TEKT5, ...*
Epidermal Cell Differentiation	GO:0009913	−1.95	< 0.01	0.012	14/42	*SPRR1B, KRT16, SPRR1A, SPRR3, IVL, WNT5A, OVOL1, SPRR2F, ...*
Vitamin D Metabolic Process	GO:0042359	−1.92	< 0.01	0.025	3/7	*CYP24A1, UGT1A4, CYP1A1*
Keratinocyte Differentiation	GO:0030216	−1.89	< 0.01	0.042	12/32	*SPRR1B, KRT16, SPRR1A, SPRR3, IVL, WNT5A, SPRR2F, S100A7, ...*
Skin Development	GO:0043588	−1.88	< 0.01	0.044	16/48	*SPRR1B, KRT16, SPRR1A, SPRR3, COL3A1, IVL, COL1A2, WNT5A, ...*
Heparan Sulfate Proteoglycan Biosynthetic Process	GO:0015012	−1.87	< 0.01	0.049	9/17	*HS6ST2, NDST4, HS3ST2, HS3ST3A1, HS6ST1, EXTL1, HS3ST1, UGDH, ...*

S3.6 Gene fusions

Table S7a summarizes the 271 candidate fusion events reported by the Gene fusion Detection module, and Table S7b gives the number of events per sample. Table S8 lists the 61 events supported by ten or more reads, and Table S9 groups the 28 in-frame events among them by gene. Values were taken from the Arriba visualizations, which give for each event the partner genes, the breakpoints, the split reads at each breakpoint, the discordant mates, the predicted reading frame and the retained protein domains. All 271 events are listed in Supplementary Data S1, and the visualizations are included in Supplementary Data S2. Recurrent events and intragenic events with closely spaced breakpoints are discussed in Section IV-F of the main article.

Table S7a. Summary of the candidate fusion events retained by Arriba. Supporting reads are split reads at both breakpoints plus discordant mates. Five events are listed without supporting reads, and each of them shares a breakpoint with a supported event of the same sample.

Measure	Value
Samples analyzed	39
Samples with at least one event	38
Candidate events (IPF, control)	271 (124, 147)
Events per sample, median (range)	6 (0 to 23)
Distinct gene pairs, either orientation	186
Intrachromosomal events	191 (70.5%)
Both breakpoints within one gene	51
Predicted in frame	54
Predicted out of frame	45
No frame prediction, non-coding partner	52
No frame prediction, no protein domain retained	27
No frame prediction, other reasons	93
At least one retained protein domain	172
No retained protein domain	99
Supporting reads per event, median (range)	3 (0 to 67)
Events with ten or more supporting reads	61, of which 28 in frame

Table S7b. Candidate fusion events per sample, by the reading frame predicted by Arriba.

Run	Group	Events	In frame	Out of frame	Non-coding partner	No domain retained	Not stated	Ten or more reads
SRR5120902	IPF	6	0	1	2	3	0	1
SRR5120903	IPF	3	1	0	1	0	1	1
SRR5120904	IPF	2	0	0	2	0	0	0
SRR5120905	IPF	10	2	1	1	3	3	4
SRR5120906	IPF	10	1	3	4	0	2	4
SRR5120907	IPF	2	1	1	0	0	0	0
SRR5120908	IPF	3	1	0	1	0	1	2
SRR5120909	IPF	10	0	1	2	0	7	0
SRR5120910	IPF	4	0	0	0	0	4	0
SRR5120911	IPF	12	4	1	0	3	4	4
SRR5120912	IPF	7	1	0	0	0	6	4
SRR5120913	IPF	9	1	1	0	1	6	1
SRR5120914	IPF	5	2	1	0	0	2	1
SRR5120915	IPF	3	1	1	1	0	0	0
SRR5120916	IPF	7	2	2	0	0	3	3
SRR5120917	IPF	11	2	0	5	1	3	4
SRR5120918	IPF	3	1	0	0	0	2	1
SRR5120919	IPF	6	3	0	0	0	3	3
SRR5120920	IPF	4	1	2	0	0	1	1
SRR5120921	IPF	7	1	0	3	2	1	1
SRR5120922	Control	0	0	0	0	0	0	0
SRR5120923	Control	3	0	1	0	1	1	0
SRR5120924	Control	9	1	2	0	0	6	4
SRR5120925	Control	3	1	0	0	0	2	0
SRR5120926	Control	6	1	1	0	0	4	2
SRR5120927	Control	4	1	1	1	0	1	1
SRR5120928	Control	15	4	5	1	2	3	3
SRR5120929	Control	3	1	0	0	0	2	0
SRR5120930	Control	10	1	0	3	2	4	2
SRR5120931	Control	5	3	1	0	1	0	2
SRR5120932	Control	4	1	1	1	0	1	0
SRR5120933	Control	16	2	2	8	0	4	2
SRR5120934	Control	10	5	4	0	0	1	1
SRR5120935	Control	6	1	0	2	0	3	1
SRR5120936	Control	23	1	7	6	6	3	2
SRR5120937	Control	8	1	1	4	1	1	4
SRR5120938	Control	5	2	0	0	0	3	1
SRR5120939	Control	6	2	2	1	1	0	1
SRR5120940	Control	11	1	2	3	0	5	0

Table S8. Candidate fusion events supported by ten or more reads, ordered by total support. Split reads are given for the first and second breakpoints. Breakpoints are hg38 coordinates. Partners given as Ensembl identifiers had no gene symbol in the annotation, and intergenic breakpoints are given as the two neighboring genes with their distances in base pairs. Discordant, discordant mates. Non-coding, one partner is non-coding. No domain, no protein domain retained.

Run	Group	Gene 1	Gene 2	Breakpoint 1	Breakpoint 2	Split reads	Discordant	Total	Reading frame
SRR5120911	IPF	*MCCC1*	*IGKC*	chr3:183114981	chr2:88860920	0, 66	1	67	Not stated
SRR5120911	IPF	*ENSG00000280441*	*IGHG1*	chr21:8399806	chr14:105741922	1, 40	21	62	No domain
SRR5120930	Control	*GTF2A2*	*GTF2A2*	chr15:59639132	chr15:59639144	31, 25	0	56	Not stated

Run	Group	Gene 1	Gene 2	Breakpoint 1	Breakpoint 2	Split reads	Discordant	Total	Reading frame
SRR5120911	IPF	*ENSG00000281181*	*IGHG1*	chr21:8438037	chr14:105742177	2, 34	18	54	No domain
SRR5120924	Control	*SFTPB*	*ENSG00000226861 (29072), ENSG00000226990 (70121)*	chr2:85658053	chr10:8330739	49, 1	0	50	Not stated
SRR5120916	IPF	*IGF1R*	*TTC23*	chr15:98891637	chr15:99218469	11, 26	9	46	Out of frame
SRR5120911	IPF	*ENSG00000280441*	*SCGB1A1*	chr21:8392869	chr11:62419087	6, 7	25	38	Not stated
SRR5120917	IPF	*ENSG00000280800*	*SFTPA1*	chr21:8210954	chr10:79614900	13, 18	0	31	Not stated
SRR5120903	IPF	*ACIN1*	*ACIN1*	chr14:23079590	chr14:23079595	14, 16	0	30	In frame
SRR5120937	Control	*ENSG00000290823*	*LINC01237*	chr15:101966223	chr2:242051920	6, 24	0	30	Non-coding
SRR5120937	Control	*ENSG00000290823*	*LINC01880*	chr15:101966223	chr2:242051920	6, 24	0	30	Non-coding
SRR5120937	Control	*WASH3P*	*LINC01237*	chr15:101966223	chr2:242051920	5, 24	0	29	Non-coding
SRR5120937	Control	*WASH3P*	*LINC01880*	chr15:101966223	chr2:242051920	5, 24	0	29	Non-coding
SRR5120924	Control	*SFTPB*	*RTP3 (7683), LRRC2 (6752)*	chr2:85658041	chr3:46508633	8, 17	3	28	Not stated
SRR5120926	Control	*ACIN1*	*ACIN1*	chr14:23079590	chr14:23079595	6, 19	0	25	In frame
SRR5120939	Control	*ACIN1*	*ACIN1*	chr14:23079590	chr14:23079595	9, 16	0	25	In frame
SRR5120928	Control	*NIPA2*	*NIPA2*	chr15:22866767	chr15:22866765	16, 8	0	24	In frame
SRR5120902	IPF	*IGLV1-47*	*CNNM4*	chr22:22358251	chr2:96807033	23, 0	0	23	No domain
SRR5120928	Control	*LMO7*	*LMO7*	chr13:75835201	chr13:75835196	13, 10	0	23	In frame
SRR5120920	IPF	*ACIN1*	*ACIN1*	chr14:23079590	chr14:23079595	7, 15	0	22	In frame
SRR5120936	Control	*CUL5*	*SLC35F2*	chr11:108078240	chr11:107878816	7, 10	5	22	Out of frame
SRR5120908	IPF	*ICAM1*	*ENSG00000282885*	chr19:10286452	chr14:49862798	11, 7	3	21	Not stated
SRR5120908	IPF	*ENSG00000282885*	*ICAM4-AS1*	chr14:49862798	chr19:10286452	7, 11	3	21	Non-coding
SRR5120912	IPF	*ENSG00000273259*	*DHRS4L1*	chr14:94624248	chr14:23973250	18, 3	0	21	Not stated
SRR5120912	IPF	*ENSG00000273259*	*DHRS4L2*	chr14:94624248	chr14:23973250	18, 3	0	21	Not stated
SRR5120912	IPF	*ENSG00000273259*	*DHRS4-AS1*	chr14:94624248	chr14:23973250	18, 3	0	21	Not stated
SRR5120935	Control	*ACIN1*	*ACIN1*	chr14:23079590	chr14:23079595	11, 8	0	19	In frame
SRR5120905	IPF	*ICAM1*	*ENSG00000282885*	chr19:10286452	chr14:49862798	5, 7	5	17	Not stated
SRR5120905	IPF	*ENSG00000282885*	*ICAM4-AS1*	chr14:49862798	chr19:10286452	7, 5	5	17	Non-coding
SRR5120916	IPF	*PRUNE2*	*PRUNE2*	chr9:76709219	chr9:76709230	7, 10	0	17	In frame
SRR5120917	IPF	*TRMO*	*TRMO*	chr9:97904746	chr9:97904754	8, 9	0	17	In frame
SRR5120924	Control	*ACIN1*	*ACIN1*	chr14:23079590	chr14:23079595	4, 12	0	16	In frame
SRR5120930	Control	*ACIN1*	*ACIN1*	chr14:23079590	chr14:23079595	7, 9	0	16	In frame
SRR5120934	Control	*ENSG00000124593*	*ENSG00000124593*	chr6:41786259	chr6:41786251	9, 7	0	16	In frame
SRR5120906	IPF	*ENSG00000247765*	*ENSG00000280881*	chr15:21324988	chr15:21330231	6, 5	4	15	Non-coding
SRR5120917	IPF	*SFTPA2*	*ENSG00000280441*	chr10:79557089	chr21:8393305	6, 0	9	15	Not stated
SRR5120921	IPF	*IST1*	*IST1*	chr16:71922614	chr16:71922609	10, 4	0	14	In frame
SRR5120933	Control	*TCIRG1*	*TCIRG1*	chr11:68050530	chr11:68050525	11, 3	0	14	In frame
SRR5120905	IPF	*ACIN1*	*ACIN1*	chr14:23079590	chr14:23079595	2, 11	0	13	In frame
SRR5120906	IPF	*CIMIP2B*	*CIMIP2B*	chr9:35562400	chr9:35562403	2, 11	0	13	In frame
SRR5120912	IPF	*ACIN1*	*ACIN1*	chr14:23079590	chr14:23079595	4, 8	0	12	In frame
SRR5120913	IPF	*ACIN1*	*ACIN1*	chr14:23079590	chr14:23079595	3, 9	0	12	In frame
SRR5120914	IPF	*ACIN1*	*ACIN1*	chr14:23079590	chr14:23079595	4, 8	0	12	In frame
SRR5120919	IPF	*FHIP2B*	*RNA5SP528 (163950), DEFB115 (277440)*	chr8:22104277	chr20:30980224	3, 9	0	12	Not stated
SRR5120919	IPF	*ACIN1*	*ACIN1*	chr14:23079590	chr14:23079595	4, 8	0	12	In frame
SRR5120931	Control	*ACIN1*	*ACIN1*	chr14:23079590	chr14:23079595	6, 6	0	12	In frame
SRR5120906	IPF	*ENSG00000282885*	*ENSG00000261759*	chr14:49862787	chr16:19120163	5, 4	2	11	Non-coding
SRR5120906	IPF	*ITPRIPL2*	*ENSG00000282885*	chr16:19120163	chr14:49862787	4, 5	2	11	Not stated
SRR5120916	IPF	*ACIN1*	*ACIN1*	chr14:23079590	chr14:23079595	3, 8	0	11	In frame
SRR5120917	IPF	*ACIN1*	*ACIN1*	chr14:23079590	chr14:23079595	2, 9	0	11	In frame
SRR5120918	IPF	*PRUNE2*	*PRUNE2*	chr9:76709219	chr9:76709230	3, 8	0	11	In frame
SRR5120919	IPF	*IST1*	*IST1*	chr16:71922614	chr16:71922609	6, 5	0	11	In frame
SRR5120927	Control	*ZNF417*	*ZNF417*	chr19:57908955	chr19:57908997	4, 7	0	11	Out of frame
SRR5120905	IPF	*ENSG00000280441*	*IGHG1*	chr21:8398893	chr14:105742353	0, 4	6	10	No domain
SRR5120924	Control	*CTIF*	*COX16*	chr18:48761693	chr14:70359673	5, 5	0	10	Out of frame

Run	Group	Gene 1	Gene 2	Breakpoint 1	Breakpoint 2	Split reads	Discordant	Total	Reading frame
SRR5120926	Control	*LAMB2*	*LAMB2*	chr3:49126114	chr3:49126159	1, 9	0	10	Out of frame
SRR5120928	Control	*SGMS2*	*SLC2A9*	chr4:107911819	chr4:9845736	6, 2	2	10	Not stated
SRR5120931	Control	*CIMIP2B*	*CIMIP2B*	chr9:35562400	chr9:35562403	6, 4	0	10	In frame
SRR5120933	Control	*LAMB2*	*LAMB2*	chr3:49126114	chr3:49126159	0, 10	0	10	Out of frame
SRR5120936	Control	*ACIN1*	*ACIN1*	chr14:23079590	chr14:23079595	2, 8	0	10	In frame
SRR5120938	Control	*ACIN1*	*ACIN1*	chr14:23079590	chr14:23079595	3, 7	0	10	In frame

Table S9. In-frame fusion candidates supported by ten or more reads (split reads plus discordant mates), grouped by gene. All were intragenic. Retained protein domains are those reported by Arriba for the predicted product, and breakpoints are given in hg38 coordinates.

Gene	Breakpoints	Distance (bp)	Samples (IPF, control)	Most supporting reads	Retained protein domains
ACIN1	chr14:23079590 and 23079595	5	9, 8	30	SAP domain, RNSP1-SAP18 binding (RSB) motif
PRUNE2	chr9:76709219 and 76709230	11	2, 0	17	DHHA2 domain, Divergent CRAL/TRIO domain, Bcl2-/adenovirus E1B nineteen kDa-interacting protein 2
IST1	chr16:71922614 and 71922609	5	2, 0	14	Regulator of Vps4 activity in the MVB pathway
CIMIP2B	chr9:35562400 and 35562403	3	1, 1	13	Protein of unknown function (DUF2475), Protein of unknown function (DUF2475)
NIPA2	chr15:22866767 and 22866765	2	0, 1	24	Magnesium transporter NIPA
LMO7	chr13:75835201 and 75835196	5	0, 1	23	Calponin homology (CH) domain, Domain of unknown function (DUF4757), PDZ domain, LIM domain
TRMO	chr9:97904746 and 97904754	8	1, 0	17	tRNA-methyltransferase O
ENSG00000124593	chr6:41786259 and 41786251	8	0, 1	16	PET Domain, LIM domain, LIM domain
TCIRG1	chr11:68050530 and 68050525	5	0, 1	14	V-type ATPase 116kDa subunit family, V-type ATPase 116kDa subunit family

S3.7 Supporting figures

Figs. S1 to S5 show further outputs of the platform for the case study. Fig. S1 compares read quality before and after trimming, Fig. S2 shows the diagnostics returned by the Gene Splicing module, and Fig. S3 shows additional outputs of the two optional modules. Fig. S4 collects the outputs of the Quantification and DEG analysis modules that are not shown in the main article, and Fig. S5 shows the top-ranked differential junctions.

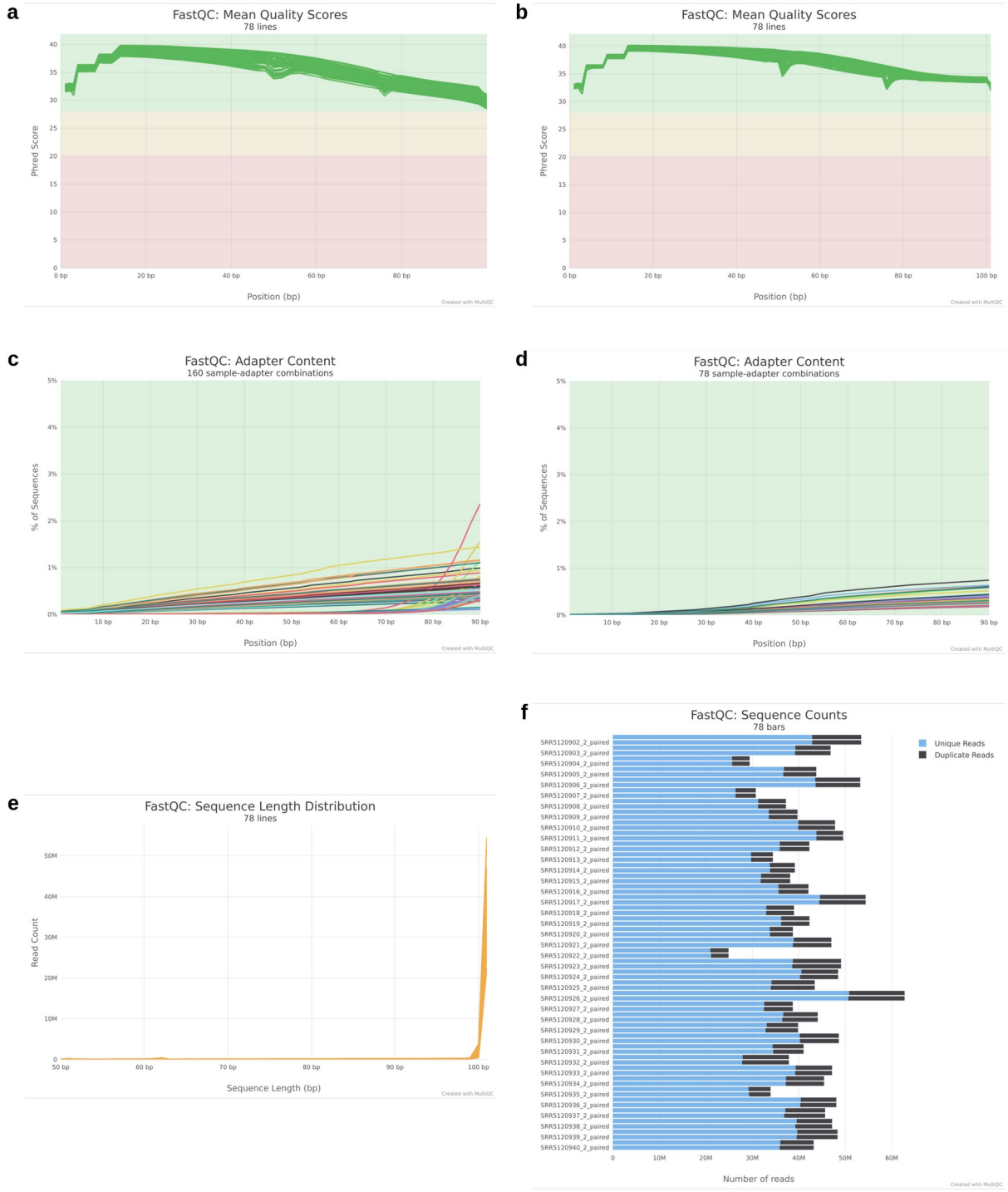


Fig. S1. Read quality before and after trimming, from the MultiQC reports of the Pre-trim QC and Post-trim QC modules. (a, b) Mean quality score along the read for the 78 raw files (a) and the 78 trimmed files (b). (c, d) Adapter content along the read before (c) and after (d) trimming. (e) Length distribution of the trimmed reads. (f) Number of sequences in each trimmed file, divided into unique and duplicate reads.

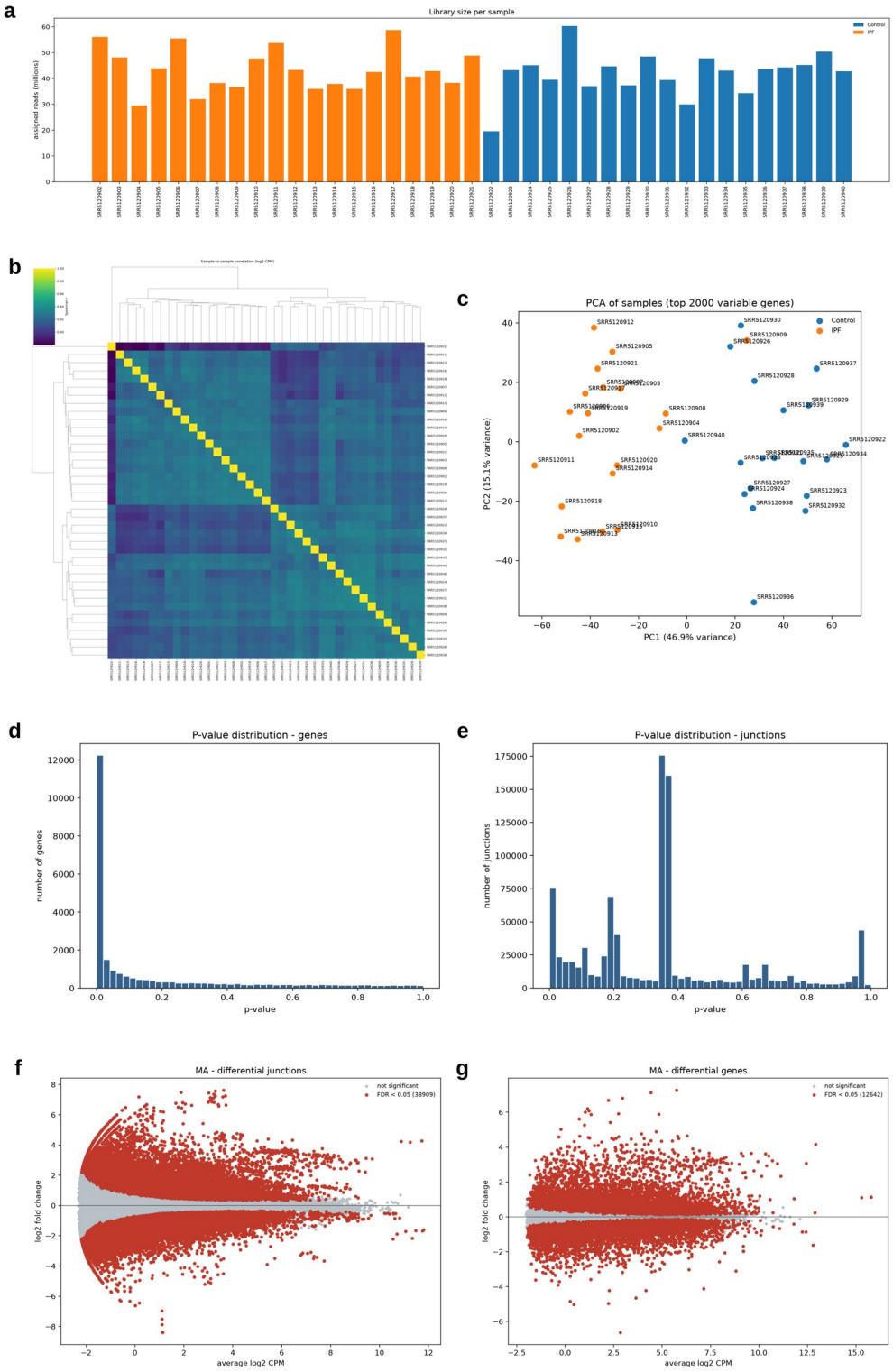


Fig. S2. Diagnostics from the splicekit report of the Gene Splicing module. (a) Library size per sample. (b) Sample-to-sample correlation. (c) Principal component analysis of the samples. (d, e) Distribution of P values for gene-level features (d) and junctions (e). (f, g) MA plots of differential junctions (f) and gene-level features (g).

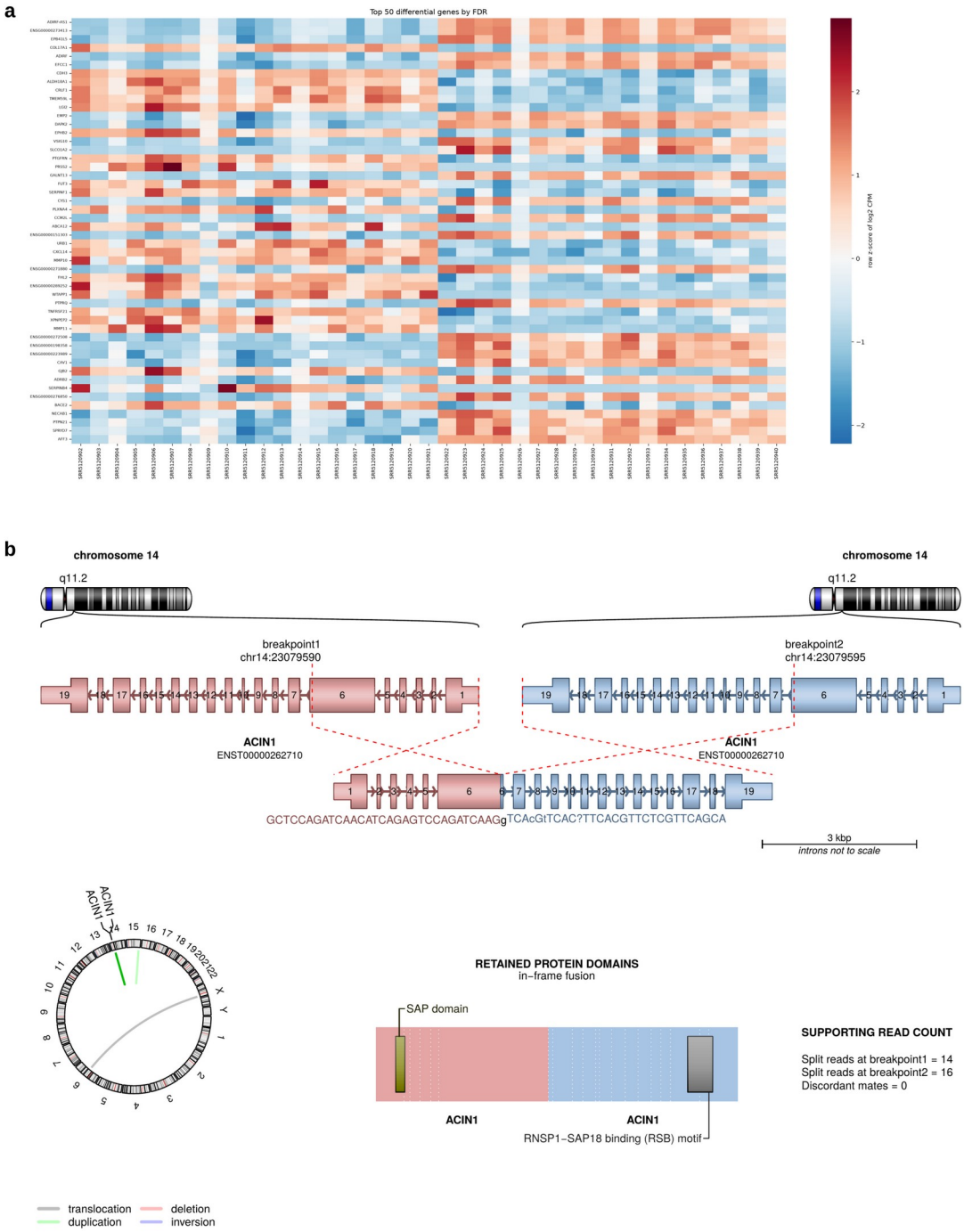


Fig. S3. Additional outputs of the optional modules. (a) Row-scaled values of the 50 most significant gene-level features across the 39 samples, from the splicekit report. (b) Arriba visualization of the recurrent intragenic *ACIN1* event in SRR5120903, one of the 19 samples in which it was found. The breakpoints lie 5 bp apart, the event is predicted to be in frame and 30 reads support it.

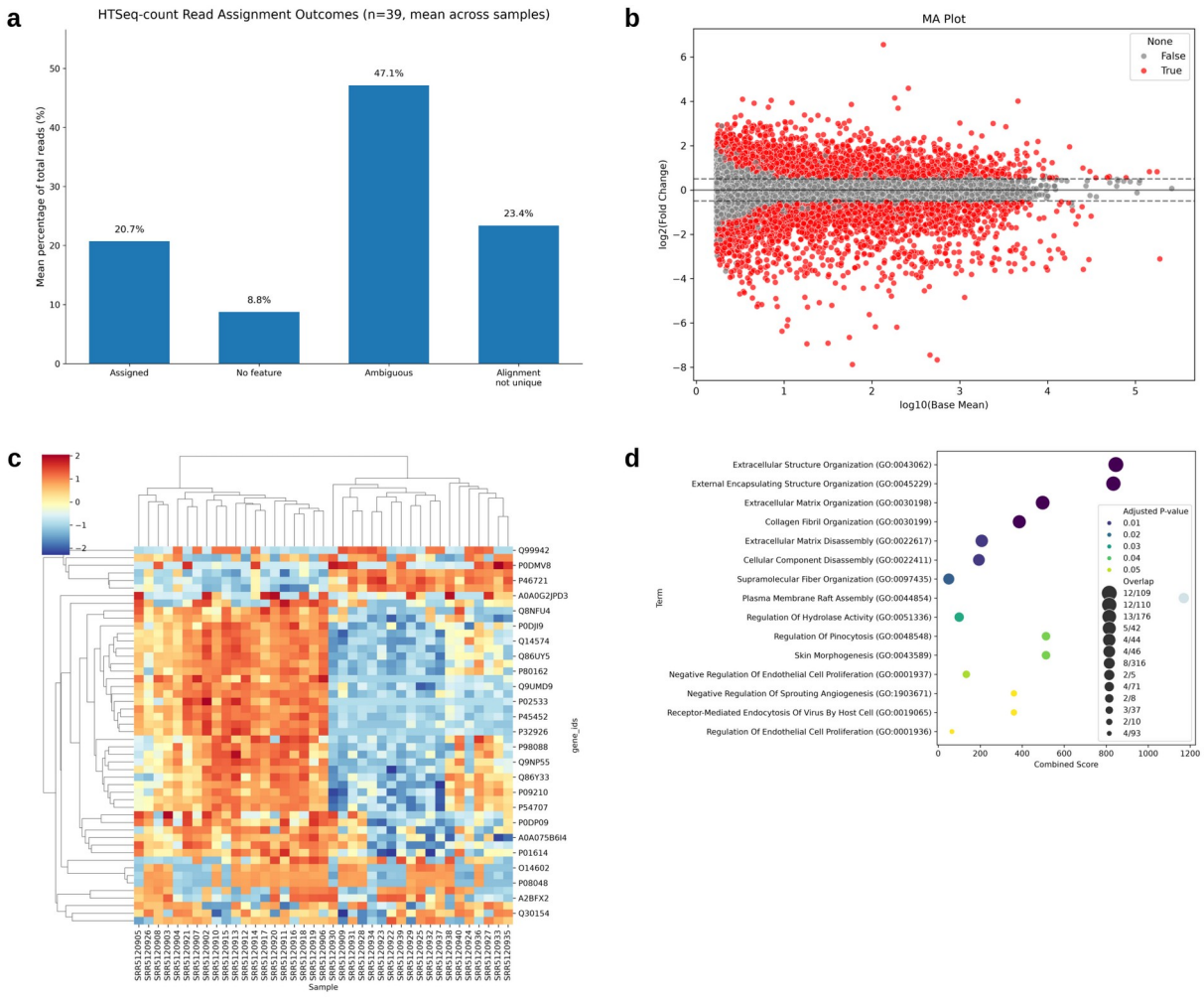


Fig. S4. Further outputs of the Quantification and DEG analysis modules. (a) Mean share of counted read pairs in each HTSeq-count category across the 39 samples. (b) MA plot of the contrast of control with IPF, with differentially expressed features in red. Fold changes are control relative to IPF, so features raised in IPF lie at negative values. (c) Clustered heatmap of row-scaled expression for the features selected by the module, labeled by UniProt accession. (d) Bubble chart of the 15 GO Biological Process terms of Fig. 4 of the main article, plotted against the Enrichr combined score, colored by adjusted P value and sized by the overlap between the input list and the term.

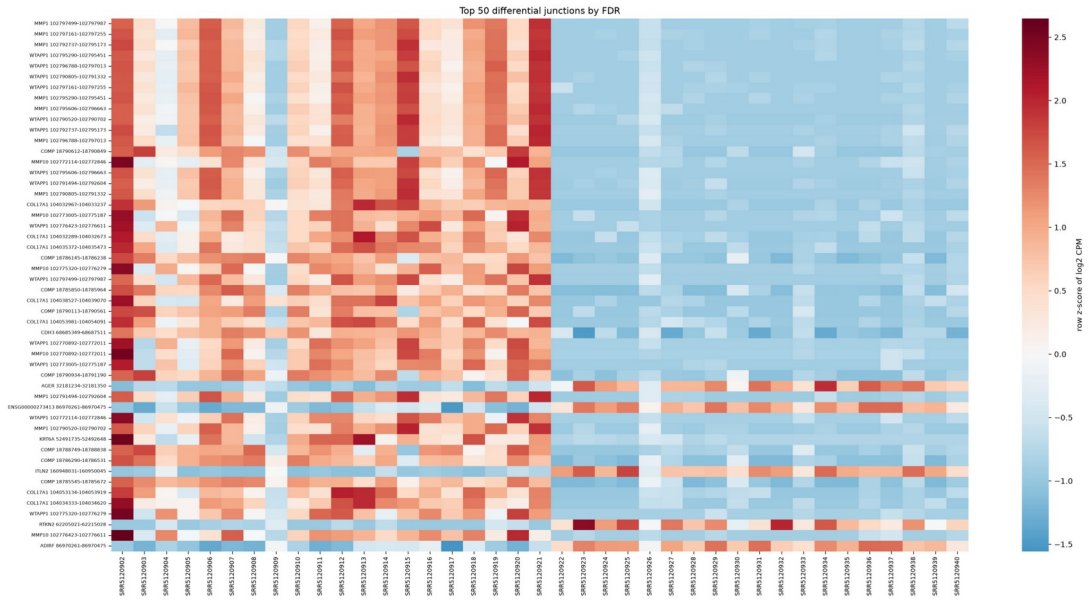


Fig. S5. Row-scaled values of the 50 most significant differential junctions across the 39 samples, from the splicekit report of the Gene Splicing module. The samples are ordered with the IPF samples (SRR5120902 to SRR5120921) first.

S4 Independent reproduction of the differential expression result

S4.1 Procedure

The annotated count matrix (annotated_counts_IPF.csv) and the sample metadata (IPF_meta_data.csv) were exported from the platform and analyzed on a separate computer with PyDESeq2 0.5.4 [14]. The 39 samples were matched to their groups by run accession. Features with no reads in any sample were removed, which left 33,877 features. The Group column of the metadata table served as the condition. The design was ~Condition, and the contrast was control against IPF, as in the platform, so fold changes were expressed as control relative to IPF. Size factors, dispersions, Wald tests and the Benjamini and Hochberg adjustment used the defaults of PyDESeq2. Features with a baseMean below 10 were then set aside, and differential expression was called at an adjusted P below 0.05 and an absolute $\log_2$ fold change above 0.5. The platform's result table was opened only after this analysis was complete.

S4.2 Agreement with the platform

The baseMean filter left the same 10,484 features as in the platform's result table. Both analyses called the same 3,464 features differentially expressed, 1,714 up-regulated and 1,750 down-regulated, and no feature was called differently. Base means agreed to within 1.5×10^{-10}, and $\log_2$ fold changes agreed to within 1.1×10^{-4}. Adjusted P values agreed to within 0.001 for 10,473 features. For the remaining 11 features the largest difference was 0.036, and none of these differences changed a significance call. Small differences of this kind are expected from the numerical optimization used to fit dispersions. The per-feature comparison is given in Supplementary Data S1.

S5 Pseudocode of the pipeline driver

Algorithm 1 is the driver of the main RNA-seq path. It chains the stages through a single job identifier, supports partial runs that resume from a completed stage and records the outcome of every stage. Algorithm 2 gives the seven stages it invokes, which correspond to the Pre-trim QC, Trimming, Alignment, Pre-quantification, Quantification, Annotation and DEG analysis modules of Table II of the main article. Each stage invokes the same service function as the corresponding web route, with the same arguments.

Algorithm 1. RNA-seq differential expression pipeline (driver)

Input R, reads in FASTQ format, paired-end or single-end, given as a registered dataset or a directory

G, reference genome name, which resolves through the configuration to the STAR genome index I_G, the GTF annotation A_G and the gene annotation table Q_G

P, θ_{trim}, adapter file and trimming parameters (leading, trailing, sliding window, minimum length, Phred encoding, head and tail crop)

q, θ_q, quantifier $q \in$ {featureCounts, htseq-count} and its parameters (feature type, attribute, strandedness, paired-end flag)

M, (c_1, c_2), sample to condition map and contrast (both optional)

S, stages to run, $S \subseteq \langle 1, \ldots, 7 \rangle$ in pipeline order (default all)

t, keepGoing, threads per tool and whether to continue after a failed stage (default false)

Output J, Φ, stage outputs under the job directory J and run report Φ

1: j ← supplied job identifier, else "pipetest_" + timestamp

2: J ← JobDirectory(user, j) ▷ *one directory, from which every stage reads its input*

3: bound the per-process thread pools (BLAS, OpenMP, joblib) ▷ *concurrent jobs must not oversubscribe the server*

4: **if** R is an ad hoc directory **then** register R as a dataset so that stage 1 accepts it

5: open the run log in J, $\Phi \leftarrow \langle$j, R, G, t, S, start time, steps = [ ]$\rangle$

6: **for each** stage s $\in$ S in pipeline order **do**

7: $t_0 \leftarrow$ now

8: **try**

9: (result, summary) $\leftarrow$ RunStage(s) $\triangleright$ *Algorithm 2*

10: **if** result reports an error or status = FAILED **then raise** error

11: append $\langle$s, OK, summary, now $- t_0\rangle$ to Φ.steps

12: **catch** interruption, append $\langle$s, INTERRUPTED, now $- t_0\rangle$ to Φ.steps and **break**

13: **catch** error e, append $\langle$s, FAILED, e, now $- t_0\rangle$ to Φ.steps

14: **if** the last entry of Φ.steps is not OK and ¬keepGoing **then break**

15: Φ.status $\leftarrow$ COMPLETED, FAILED or INTERRUPTED, and record the total run time

16: print the per-stage summary and write Φ as JSON to J

17: **return** Φ

Algorithm 2. RunStage(s), the pipeline stages

Each stage invokes the same service function as the corresponding web route, with the same arguments. require X aborts the stage, which Algorithm 1 then reports as FAILED, if X does not hold.

Stage 1, FastQC, read quality control

1: **for each** sample r $\in$ R **do** FastQC(r) $\rightarrow$ QC reports and plots in J/fastqc

2: **return** $\langle$#samples, #paired-end, #single-end, #succeeded, #failed, #plots$\rangle$

Stage 2, Trimmomatic, adapter and quality trimming

3: **require** adapter file P exists

4: **for each** sample r $\in$ R, resolved through job j, **do** Trimmomatic(r, P, θ_{trim}, t) $\rightarrow$ trimmed FASTQ in J/trimmomatic

5: **return** $\langle$adapter, #samples, #succeeded, #failed, mean read survival rate$\rangle$

Stage 3, STAR, spliced read alignment

6: **for each** trimmed sample r′ of job j **do** STAR(r′, I_G, A_G, t) $\rightarrow$ aligned BAM in J/star/alignment

7: **return** $\langle$reference, #samples, #succeeded, #failed, overall alignment rate$\rangle$

Stage 4, pre-quantification BAM processing

8: D $\leftarrow$ J/star/alignment

9: B $\leftarrow$ BAM files in D, **require** B $\neq \varnothing$

10: **for each** b $\in$ B **do** sort $\rightarrow$ fixmate $\rightarrow$ mark duplicates $\rightarrow$ index $\rightarrow$ b_markdup.bam in J/sorting

11: **return** $\langle$aligner, |B|, #succeeded, #failed, duplicate rate$\rangle$

Stage 5, quantification, count matrix

12: **require** A_G configured and present, B′ $\leftarrow$ *_markdup.bam files in J/sorting, **require** B′ $\neq \varnothing$

13: **if** q = featureCounts **then**

14: featureCounts(B', A_G, θ_q, t) ▷ *one multithreaded run over all samples, with the GTF parsed once*

15: **else**

16: **for each** b ∈ B' **do** htseq-count(b, A_G, θ_q) ▷ *one single-threaded run per sample*

17: merge the per-sample counts → count matrix C (genes × samples), written as merged_counts.csv

18: **return** ⟨quantifier, A_G, #samples, #succeeded, assignment rate, #genes⟩

Stage 6, gene annotation

19: **require** Q_G present, C ← merged_counts.csv of job j, from the directory of either quantifier

20: C_{ann} ← join C with Q_G on gene identifier → annotated_counts_j.csv in J/annotation

21: **return** ⟨#annotated genes, #unannotated genes, annotation rate⟩

Stage 7, DESeq2, differential expression

22: C ← C_{ann} if present, else merged_counts.csv

23: **if** M not supplied **then** ▷ *default two-group design*

24: samples ← sample columns of C, sorted by name, **require** |samples| ≥ 2

25: M ← first ⌊|samples| / 2⌋ samples ↦ c_A, remaining samples ↦ c_B

26: conditions ← distinct values of M, **require** |conditions| ≥ 2

27: (c_1, c_2) ← supplied contrast, else the first two conditions

28: DESeq2(C, M, (c_1, c_2)) → results in J/deseq2

29: **return** ⟨count file, contrast, M, #genes tested, #significant DEGs, #up-regulated, #down-regulated⟩

In this notation the fold change reported by stage 7 is expressed as c_1 relative to c_2. In the case study c_1 was control and c_2 was IPF, so the exported fold changes, and those reported in the main article, are expressed as control relative to IPF.

S6 Supplementary Data S1 and S2

Supplementary Data S1 (OmicsXplore_Data_S1.xlsx) is a workbook of ten sheets, and its README sheet describes every sheet and column. Samples and Per_sample_metrics hold the values of Tables S2 and S3. DE_all_tested holds the result for all 10,484 tested features and DE_significant the 3,464 differentially expressed features. In both sheets fold changes and Wald statistics are given as exported, that is, as control relative to IPF, and a further column states whether each differentially expressed feature is raised or reduced in IPF. ORA_GO_BP_2023 holds all 753 terms returned by Enrichr. GSEA_prerank holds all 4,735 gene sets as exported. Fusions_all lists the 271 candidate fusion events and Fusions_per_sample the counts of Table S7b. Reproduction compares the platform's result with the independent re-run of Section S4 for every feature.

Supplementary Data S2 (OmicsXplore_Data_S2.zip) is an archive of the files exported by the platform for the case study. It contains the MultiQC reports of the Pre-trim QC and Post-trim QC modules and of the full pipeline, the SAMtools flagstat table, the HTSeq-count statistics, the annotated count matrix, the sample metadata and the SRA run table. It also holds all tables and plots of the DEG analysis module, the splicekit report and the 38 Arriba PDF files that contain the 271 fusion visualizations. A README file in the archive describes each file and the orientation of the exported fold changes.

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